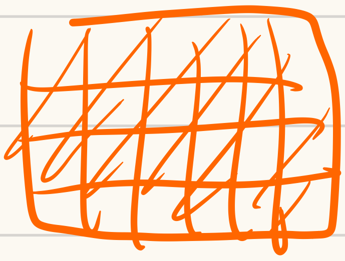


矩阵理论

习题课

19 Oct 2026

$$Ax = b$$

$$(\alpha_1|\alpha_2|\cdots|\alpha_n)\cdot\begin{pmatrix}x_1\\ \vdots\\ x_n\end{pmatrix}$$

x 表示 A 拉伸系数 去组.

用什么样的系数去组合 A 的列
使得结果 b.
x 拉伸系数.

表示, representation

$$A = \begin{pmatrix} 2&4&1&8&-1\\ 3&6&2&13&1\\ 1&2&-2&-1&3 \end{pmatrix} \xrightarrow{约化} H_A = \begin{pmatrix} 1&2&0&3&0\\ 0&0&1&2&0\\ 0&0&0&0&1 \end{pmatrix}$$

$$\begin{array}{ccccc} \triangle & & \triangle & & \triangle \end{array}$$

row

pivot

reduced échelon form

$$H_A \begin{pmatrix} 1&2&0&3&0\\ 0&0&1&2&0\\ 0&0&0&0&1 \end{pmatrix}$$

$$a\begin{pmatrix}2\\3\\1\end{pmatrix}+b\begin{pmatrix}1\\2\\-2\end{pmatrix}=\begin{pmatrix}8\\13\\-1\end{pmatrix}$$

问题. 以上由列线性关系刻划的

$$A \longrightarrow H_A$$ 算法,

是否也是行初等变换得到的.

答: 由归纳法 (假定对 ≤k列的情形成立--)

Def. H_A 最简行阶梯形

(标准型)

设 A 是一个最简行阶梯形, 若

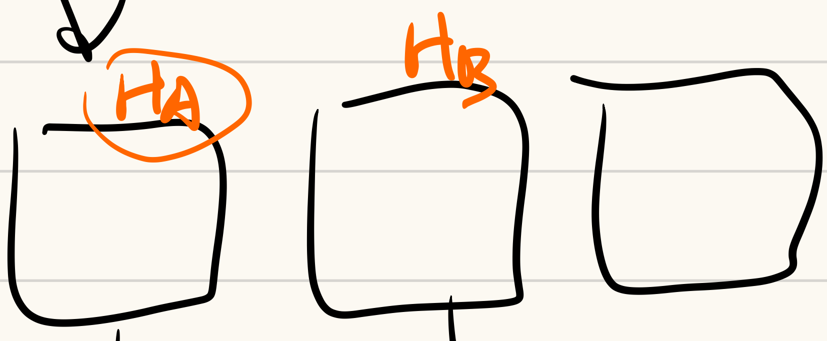
$$A = \begin{pmatrix} 1 & * & 0 & * & 0 & * & 0 & * \\ & & 1 & & 0 & & 0 & * \\ & & & & 1 & & 0 & \\ & 0 & & & & & 1 & \end{pmatrix}$$

SLOGAN. 最简行阶梯形 "分类" 了矩阵.

所有 $m\times n$ 矩阵.

$x+y=0$ $(1\ 1)$

$2x+2y=0$ $(2\ 2)$



A, A' B

$Ax=0$
$A'x=0$ 同解.

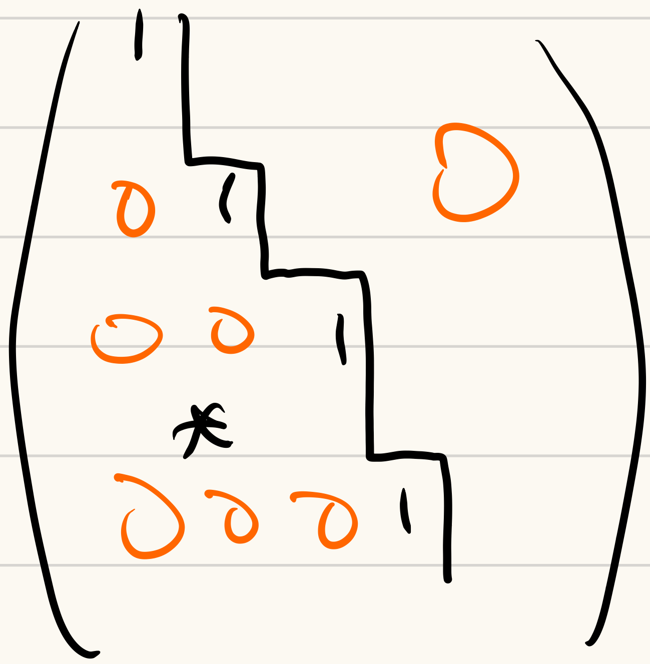
$Ax=0$
$Bx=0$
不同解.

rank、行秩

$\text{rowrank}(A)$

$= H_A$ 的非零行数

Def. 简约列阶梯形



$P\cdot(M\cdot Q)=(P\cdot M)\cdot Q$

$$\begin{tikzcd} M \arrow[r, "换 r_1, r_2"] \arrow[d, "c_1 加2倍 c_2"'] & M' \arrow[d, "c_1 加2倍 c_2"] \\ M'' \arrow[r, "换 r_1, r_2"] & M''' \end{tikzcd}$$

Thm. 行秩 = 列秩.

$\text{rowrank}(A)$

$\|$ 可逆列

rowrank (A的最简列阶梯形)

$\|$

$\text{col rank}(A)$

Ex. $\operatorname{rank}(A \mid B) \geq \operatorname{rank} A$

Ex. $\operatorname{rank}\begin{pmatrix} A \\ B \end{pmatrix} \leq \operatorname{rank} A + \operatorname{rank} B$

$$\begin{gathered} \| \\ \operatorname{rank}\begin{pmatrix} H_A \\ H_B \end{pmatrix} \leq \operatorname{rank} H_A + \operatorname{rank} H_B \end{gathered}$$

Ex. $\max(\operatorname{rank} A, \operatorname{rank} B)$

$\geq \max \sim$ 与

$$\begin{gathered} \max(\operatorname{rank} A, \operatorname{rank} B) \\ \wedge| \\ \operatorname{rank}(A \mid B) \\ \wedge| \\ \operatorname{rank} A + \operatorname{rank} B. \end{gathered}$$

✓

Ex. $\operatorname{rank}(A \cdot B) \leq \operatorname{rank}(B)$

Ex. $\operatorname{rank}\begin{pmatrix} A & B \\ 0 & C \end{pmatrix} \geq \operatorname{rank} A + \operatorname{rank} C$ ✓

研究生学了 $A \in \mathbb{R}^{m \times n}$

$$Ax = 0 \Longleftrightarrow A^T A x = 0$$

$$\Downarrow$$

$$A^T A x = 0 \Rightarrow x^T A^T A x = 0$$

$$\|Ax\|^2 \overset{①}{=} 0$$

$\{$

内积， e.g. $\boxed{\frac{2}{\pi}\int_0^{\pi} ? \cdot ? \; d\theta}$

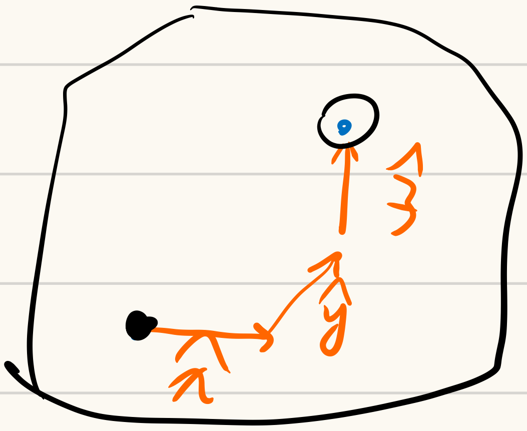
S

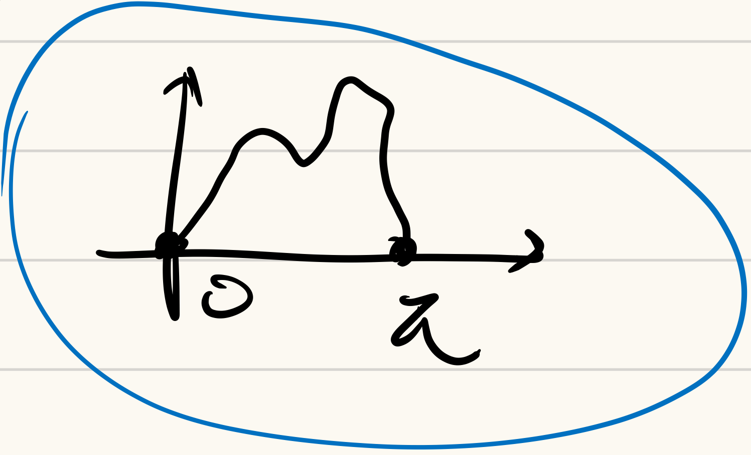
经典 $\{\sin x, \sin 2x, \sin 3x, \cdots\}$

$\boxed{} : S \times S \longrightarrow \{0, 1\}$.

$\mathbb{R}^3$ 中的内积、 $\mathbb{R}^3 \times \mathbb{R}^3 \longrightarrow \mathbb{R}$

$$\begin{pmatrix}x\\y\\z\end{pmatrix}, \begin{pmatrix}a\\b\\c\end{pmatrix} \longmapsto ax+by+cz$$





$$f \longmapsto \left(\boxed{f, \sin x}, \boxed{f, \sin 2x}, \boxed{f, \sin 3x} \ldots \right)$$

$$\boxed{f} = \frac{2}{\pi}\int_0^{\pi} f \sin x \, dx + \frac{2}{\pi}\int_0^{\pi} f \sin 2x \, dx + \cdots$$

协方差 数据的“坐标”

$\underline{\text{orthogonal}}$ $\perp$

$\mathcal{N}$ C $A \in \mathbb{R}^{m \times n}$

$$\mathcal{N}(A) = \mathcal{N}(AA^T)$$

$$\mathcal{N}(P) + \mathcal{N}(Q) = \mathcal{N}(PP^T) + \mathcal{N}(QQ^T)$$

$$= \mathcal{N}(PP^T + QQ^T)$$

$$= \mathcal{N}\left(\begin{pmatrix} P & Q \end{pmatrix} \begin{pmatrix} P^T \\ Q^T \end{pmatrix}\right)$$

$$= \mathcal{N}\begin{pmatrix} P & Q \end{pmatrix}$$

$$= \mathcal{N}$$

$$R(P+Q) = R(PP^T + QQ^T)$$

$$= R\left((P|Q) \cdot (P|Q)^T\right)$$

$$= R(P|Q)$$

$$= \underline{R(P)} + \underline{R(Q)}$$

$$\begin{tikzcd}
B \arrow[r, "U_1"] & \begin{pmatrix}\lambda_1 & 0 \\ 0 & B_1\end{pmatrix} \arrow[r, "\begin{pmatrix}1 & 0 \\ 0 & U_2\end{pmatrix}"] & \begin{pmatrix}\lambda_1 & & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & B_2\end{pmatrix} \arrow[d] \\
& & \vdots \\
& & \begin{pmatrix}\lambda & & \\ & \ddots & \\ & & \lambda\end{pmatrix}
\end{tikzcd}$$